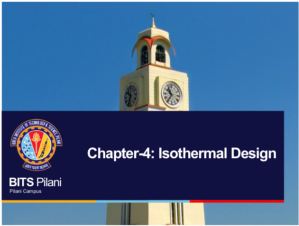




1



2

Outline

- Describe the algorithm that allows the reader to solve chemical reaction engineering problems through logic rather than memorization.
- Sizing the batch reactors, CSTRs, PFRs, and PBRs for isothermal operation given the rate law and feed conditions.
- Account for the effects of pressure drop on conversion in packed bed tubular reactors.

3

Review Chapter 1

The GMBE applied to the four major reactor types (and the general reaction $A \rightarrow B$)

Reactor	Differential	Algebraic	Integral
Batch	$\frac{dN_A}{dt} = r_A V$		$t = \int_{N_{A0}}^{N_A} \frac{dN_A}{r_A V}$ 
CSTR		$V = \frac{F_{A0} - F_A}{-r_A}$	
PFR	$\frac{dF_A}{dV} = r_A$		$V = \int_{F_{A0}}^{F_A} \frac{dF_A}{r_A}$ 
PBR		$\frac{dF_A}{dW} = r_A'$	$W = \int_{F_{A0}}^{F_A} \frac{dF_A}{r_A'}$ 

4

29-09-2023

ONE 211 Kinetics & Reactor Design

Review Chapter 2

In terms of Conversion

Reactor Differential Algebraic Integral

Batch $N_{A0} \frac{dX}{dt} = -r_A V$

Integral $t = N_{A0} \int_0^X \frac{dX}{-r_A V}$

CSTR $V = \frac{F_{A0} X}{-r_A}$

PFR $F_{A0} \frac{dX}{dV} = -r_A$

Integral $V = F_{A0} \int_0^X \frac{dX}{-r_A}$

8 (29 September 2023)

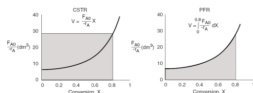
MTS Plots, Plot Compare

5

ONE 211 Kinetics & Reactor Design

Review Chapter 2

Levenspiel Plots



29 September 2023

MTS Plots, Plot Compare

6

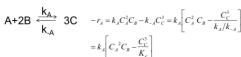
ONE 211 Kinetics & Reactor Design

Review Chapter 3

Power Law Model:

$-r_A = k C_A^\alpha C_B^\beta$
 α order in A
 β order in B
Overall Reaction Order = $\alpha + \beta$

Reversible Elementary Reaction



7 (29 September 2023)

MTS Plots, Plot Compare

7

ONE 211 Kinetics & Reactor Design

Review Chapter 3

Arrhenius Equation

$\ln k_A = \ln A - \frac{E}{R(T)}$



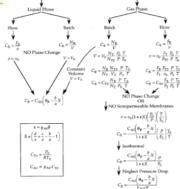
29 September 2023

MTS Plots, Plot Compare

8

CHE 311 Kinetics & Reactor Design

$$A + \frac{1}{2} B \rightarrow C + \frac{1}{2} D$$



29 September 2023

9

CHE 311 Kinetics & Reactor Design

Review 1, 2, & 3

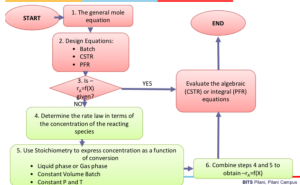


29 September 2023

10

CHE 311 Kinetics & Reactor Design

Algorithm for Isothermal Reactor



11

CHE 311 Kinetics & Reactor Design

Design of Reactor

• To design an isothermal reactors, the following sequence is highly recommended

- Mole balance and design equation
- Rate law
- Stoichiometry
- combine
- Evaluate

29 September 2023

12

Scale up of Batch Reactor to the design of CSTR

- Calculation of time taken to achieve a given conversion X 

- Step 1: Write the mole balance

- Step 2: Write the rate law

$$-r_A = k C_A^2$$

Example for second order reaction

- Step 3: Derive concentration term from stoichiometry

13



- Step 1: Write the mole balance

$$N_{A0} \frac{dX}{dt} = -r_A V$$

- Step 2: Write the rate law

$$-r_A = k C_A^2$$

Example for second order reaction

- Step 3: Derive concentration term from stoichiometry

$$C_A = C_{A0}(1-X)$$

14



- Step 4: Combine equation from step 1,2,3

$$\frac{dX}{dt} = k C_{A0}(1-X)^2$$

- Step 5: Evaluate

$$\int_0^X dt = \frac{1}{k C_{A0}} \int_0^X \frac{dX}{(1-X)^2}$$

$$t = \frac{1}{k C_{A0}} \left(\frac{X}{1-X} \right)$$

This is the reaction time or t_R

15

Algorithm to estimate reaction time

Mole Balance	$N_{A0} \frac{dX}{dt} = -r_A V$	
Rate Law	First order $-r_A = k C_A$	Second order $-r_A = k C_A^2$
Stoichiometry	$C_A = \frac{N_{A0}}{V_0} (1-X)$	
Combine	$\frac{dX}{dt} = k(1-X)$	$\frac{dX}{dt} = k C_{A0}(1-X)^2$
Evaluate (integrate)	$t_R = \frac{1}{k} \ln \left(\frac{1}{1-X} \right)$	$t_R = \frac{1}{k C_{A0}} \left(\frac{X}{1-X} \right)$

16



To reach 90% conversion in a constant-volume batch reactor scales:

$$t_R = \frac{1}{k} \ln \left(\frac{1}{1-X} \right)$$

For first order

$$= \frac{1}{k} \ln \left(\frac{1}{1-0.9} \right)$$

$$\text{if } k = 10^{-4} \text{ s}^{-1}$$

$$t_R = \frac{2.3}{10^{-4} \text{ s}^{-1}} = 23000 \text{ s} = 6.4 \text{ h}$$

29 September 2023

BUTS Planet, Planet Campus

17



Batch Reactor Data

- In the laboratory experiment, 500 mL of a 2 M solution of EO in water was mixed with 500 mL of water containing 0.9 wt % sulfuric acid, which is a catalyst. The temperature was maintained at 55°C. The concentration of ethylene glycol was recorded as a function of time. Determine the rate constant at 55 °C



Time (min)	Concentration of Ethylene Glycol (kmol/m ³)
0.0	0.000
0.5	0.145
1.0	0.270
1.5	0.376
2.0	0.467
3.0	0.610
4.0	0.715
6.0	0.848
10.0	0.957

$$1 \text{ kmol/m}^3 = 1 \text{ mol/dm}^3 = 1 \text{ mol/L}$$

29 September

BUTS Planet, Planet Campus

18

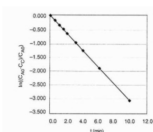


t (min)	C _C (kmol/m ³)	$\frac{C_{A0} - C_C}{C_{A0}}$	$\ln \left(\frac{C_{A0} - C_C}{C_{A0}} \right)$
0.0	0.000	1.000	0.0000
0.5	0.145	0.855	-0.1570
1.0	0.270	0.730	-0.3150
1.5	0.376	0.624	-0.4720
2.0	0.467	0.533	-0.6290
3.0	0.610	0.390	-0.9420
4.0	0.715	0.285	-1.2550
6.0	0.848	0.152	-1.8940
10.0	0.957	0.043	-3.1470

29 September 2023

BUTS Planet, Planet Campus

19



Slope, $k = 0.311 \text{ min}^{-1}$

29 September 2023

BUTS Planet, Planet Campus

20

Design of CSTR

Step 1: Write the mole balance of CSTR

Design equation for CSTR is $V = \frac{F_A X}{(-r_A)_{\text{exit}}} = \frac{v_0 C_{A0} X}{(-r_A)_{\text{exit}}}$

If volumetric flow rate does not change with the reaction, (i.e. $v = v_0$), then

$$\tau = \frac{V}{v_0} = \frac{v_0 C_{A0} X}{-r_A} = \frac{C_{A0} X}{-r_A}$$

where τ is the space time

29 September 2023

BUTS Place, Pinn Campus

21

Step 2: Write the rate law

- For 1st order irreversible reaction, $-r_A = k C_A$

Step 3: Derive concentration in terms of conversion (from stoichiometry)

$$C_A = C_{A0}(1 - X)$$

Step 4: Combine eq from step 1, 2, 3

$$\tau = \frac{1}{k} \left(\frac{X}{1 - X} \right)$$

Rearranging:

$$X = \frac{\tau k}{1 + \tau k}$$

τk is often referred to as Damköhler number (for 1st order)

29 September 2023

BUTS Place, Pinn Campus

22

Damköhler number

- Is the ratio of the rate of reaction of A to the rate of convective transport of A at the entrance to the reactor.

$$Da = \frac{-r_A V}{F_{A0}} = \frac{\text{rate of reaction at entrance}}{\text{entering flow rate of A}}$$

- For first order irreversible reaction;

$$Da = \frac{-r_A V}{F_{A0}} = \frac{k C_{A0} V}{v_0 C_{A0}} = \tau k$$

- For second order irreversible reaction;

$$Da = \frac{-r_A V}{F_{A0}} = \frac{k C_{A0}^2 V}{v_0 C_{A0}} = \tau k C_{A0}$$

29 September 2023

BUTS Place, Pinn Campus

23

Degree of conversion for a CSTR?

By using Damkohler number,

$$Da = \frac{-r_A V}{F_{A0}}$$

Rule of thumb

If $Da < 0.1$, $X < 0.1$
If $Da \geq 10$, $X > 0.9$

If first degree order, $Da = \tau k$
If second degree order, $Da = \tau k C_{A0}$

29 September 2023

BUTS Place, Pinn Campus

24

CHE 311 Kinetics & Reactor Design

CSTR-1st Order-series

- For CSTRs in series, conversion as a function of the number of tanks in series:

$$X = 1 - \frac{1}{(1 + \tau k)^n}$$

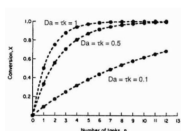
29 September 2023

BUTS Pilani, Pilani Campus

25

CHE 311 Kinetics & Reactor Design

First order system

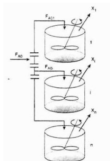


29 September 2023

BUTS Pilani, Pilani Campus

26

CHE 311 Kinetics & Reactor Design

CSTR-1st Order-parallel

$$V = \frac{F_{A0} X_1}{-r_{A1}} = \frac{F_{A0} X}{-r_A}$$

- The conversion achieved in any one of the reactors in parallel is identical to what would be achieved if the reactant were fed in one stream to one large reactor of volume V

29 September 2023

BUTS Pilani, Pilani Campus

27

CHE 311 Kinetics & Reactor Design

A Second-Order Reaction in a CSTR

$$V = \frac{F A_0 X_A}{-r_A}$$

$$X = \frac{(1 + 2kC_{A0}) - \sqrt{(1 + 2kC_{A0})^2 - (2kC_{A0})^2}}{2kC_{A0}}$$

$$= \frac{(1 + 2kC_{A0}) - \sqrt{1 + 4kC_{A0}}}{2kC_{A0}}$$

Conversion in terms of Da

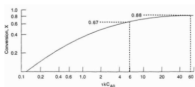
$$X = \frac{(1 + 2Da) - \sqrt{1 + 4Da}}{2Da}$$

29 September 2023

BUTS Pilani, Pilani Campus

28

CHE 311 Kinetics & Reactor Design

2nd Order reaction- CSTR

29 September 2023

BUTS Planet, Planet Campus

29

CHE 311 Kinetics & Reactor Design

Producing 200 Million Pounds per Year in a CSTR

➤ It is desired to produce 200 million pounds per year of ethylene glycol (EG). The reactor is to be operated isothermally. A 1lb mol/l³ solution of ethylene oxide (EO) in water is fed to the reactor shown in figure together with an equal volumetric solution of water containing 0.9 wt% of the catalyst H₂SO₄. The specific reaction rate constant is 0.311 min⁻¹.

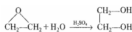
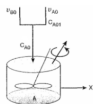
- If 80% conversion is to be achieved, determine the necessary CSTR volume.
- If 800-gal reactors were arranged in parallel, what is the corresponding conversion?
- If 800-gal reactors were arranged in series, what is the corresponding conversion?

29 September 2023

BUTS Planet, Planet Campus

30

CHE 311 Kinetics & Reactor Design



29 September 2023

BUTS Planet, Planet Campus

31

CHE 311 Kinetics & Reactor Design

Extract the given information:

$$F_C = 2 \times 10^6 \text{ lb}_m/\text{yr} \times 1 \text{ yr}/365 \text{ days} \times 1 \text{ day}/24 \text{ h} \times 1 \text{ hr}/60 \text{ min} \times 1 \text{ lbmol}/62 \text{ lb}_m = 6.137 \text{ lbmol/min}$$

From reaction stoichiometry,

$$F_C = F_{A0}X$$

$$F_{A0} = \frac{F_C}{X} = \frac{6.137}{0.8} = 7.67 \frac{\text{lbmol}}{\text{min}}$$

29 September 2023

BUTS Planet, Planet Campus

32



STEP 1: Design equation of CSTR

$$V = \frac{F_{A0}X}{(-r_A)_{\text{exit}}}$$

STEP 2: Rate Law

$$-r_A = kC_A$$

STEP 3: Stoichiometry (Liquid phase, $v = v_0$)

$$C_A = C_{A0}(1-X)$$

STEP 4: Combining

$$V = \frac{F_{A0}X}{(-r_A)_{\text{exit}}} = \frac{v_0X}{k(1-X)}$$



STEP 5: Evaluate

The entering volumetric flowrate of stream A, with $C_{A0} = 1\text{ lb mol/ft}^3$ before mixing is:

$$v_{A0} = \frac{F_{A0}}{C_{A0}} = \frac{7.67\text{ lbmol/min}}{1\text{ lbmol/ft}^3} = 7.67\frac{\text{ft}^3}{\text{min}}$$

From the problem statement, $v_{B0} = v_{A0}$

Thus, the total entering volumetric flow rate of liquid is

$$v_0 = v_{A0} + v_{B0} = 7.67 + 7.67 = 15.34\frac{\text{ft}^3}{\text{min}}$$

Substituting all the values to calculate volume of reactor:

$$V = \frac{F_{A0}X}{(-r_A)_{\text{exit}}} = \frac{v_0X}{k(1-X)} = 15.34\frac{\text{ft}^3}{\text{min}} \times \frac{0.8}{(0.311\text{ min}^{-1})(1-0.8)} = 197.3\text{ ft}^3$$



b) CSTR in parallel.

Rearranging the equation of volume in part a)

$$V = \frac{v_0X}{k(1-X)}$$

$$\frac{V}{v_0} = \frac{X}{k(1-X)}$$

$$\tau = \frac{X}{k(1-X)}$$

$$X = \frac{\tau k}{(1 + \tau k)} = \frac{Du}{(1 + Du)}$$



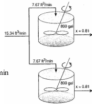
b) CSTR in parallel.

$$X = \frac{\tau k}{(1 + \tau k)} = \frac{Du}{(1 + Du)}$$

$$\tau = \frac{V}{v_0} = \frac{800\text{ gal} \times \frac{1\text{ ft}^3}{7.48\text{ gal}} \times \frac{1}{7.67\text{ ft}^3/\text{min}}}{1} = 13.94\text{ min}$$

$$Du = \tau k = 13.94\text{ min} \times 0.311\text{ min}^{-1} = 4.34$$

$$X = \frac{4.34}{(1 + 4.34)} = 0.81$$





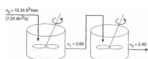
C) CSTR in series

$$X = 1 - \frac{1}{(1 + \tau k)^N}$$

$$\tau = \frac{V}{v_0} = \frac{800 \text{ gal} \times \frac{1 \text{ ft}^3}{7.48 \text{ gal}}}{15.34 \text{ ft}^3 / \text{min}} = 6.97 \text{ min}$$

$$Da = \tau k = 6.97 \text{ min} \times 0.311 \text{ min}^{-1} = 2.167$$

$$X = 1 - \frac{1}{(1 + 2.167)^2} = 0.90$$



28 September 2023

BTS: Pines, Pines Campus

37



PFR Design

- Assume no dispersion and no radial gradients in either temperature, velocity, or concentration and in the absence of pressure drop or heat exchange.

STEP 1: Write the mole balance of PFR:

$$V = F_{A0} \int_0^X \frac{dX}{-r_A}$$

STEP 2: Write the rate law

Eg: For second order,

$$-r_A = kC_A^2$$

28 September 2023

BTS: Pines, Pines Campus

38



STEP 3: Write concentration in terms of conversion (from stoichiometry)

$$C_A = C_{A0}(1 - X) \quad \text{For liquid phase}$$

$$C_A = C_{A0} \left(\frac{1 - X}{1 + \epsilon X} \right) \quad \text{For gas phase}$$

28 September 2023

BTS: Pines, Pines Campus

39

In case of 2nd order rxn,
liquid phase, isothermalNo pressure drop
No heat exchange

- mole balance $F_{A0} \frac{dX}{dV} = -r_A \quad V = F_{A0} \int_0^X \frac{dX}{-r_A}$
- rate laws $-r_A = kC_A^2$
- Stoichiometry $C_A = C_{A0}(1 - X)$
- combination $V = \frac{F_{A0}}{kC_{A0}^2} \int_0^X \frac{dX}{(1 - X)^2} = \frac{v_0 C_{A0}}{kC_{A0}^2} \left(\frac{X}{1 - X} \right)$

$$X = \frac{kC_{A0}}{1 + kC_{A0}} = \frac{Da_2}{1 + Da_2}$$

Damköhler number for 2nd-order reaction

28 September 2023

BTS: Pines, Pines Campus

41

Design a PFR: summary

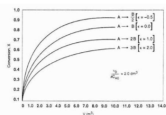
In case of 2nd order rxn, gas phase, isothermal

- mole balance $F_{A0} \frac{dX}{dV} = -r_A$ No pressure drop No heat exchange $V = F_{A0} \int_0^X \frac{dX}{-r_A}$
 - rate laws $-r_A = kC_A^2$
 - Stoichiometry $C_A = \frac{F_A}{v} = \frac{F_{A0}(1-X)}{v_0(1+\varepsilon X)} = C_{A0} \frac{(1-X)}{(1+\varepsilon X)}$
 - combination $V = F_{A0} \int_0^X \frac{(1+\varepsilon X)^2}{kC_{A0}^2(1-X)^2} dX$
- $$V = \frac{V_0}{kC_{A0}} \left[2\varepsilon(1+\varepsilon)\ln(1-X) + \varepsilon^2 X + \frac{(1+\varepsilon)^2 X}{1-X} \right]$$

29 September 2023

BUTS Planet, Planet Campus

42



29 September 2023

BUTS Planet, Planet Campus

43

PFR design

- Determine the plug-flow reactor volume necessary to produce 300 million pounds of ethylene in a year from cracking a feed stream of pure ethane. The reaction is irreversible and follows an elementary rate law. We want to achieve 80% conversion of ethane. Operating the reactor isothermally at 1100 K at a pressure of 6 atm. $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82 \text{ kCal/g.mol}$.
- Determine the No. of PFRs, if the bundle of PFRs are arranged in parallel. (one reactor volume is 0.0205 ft^3 and length is 40 ft)

29 September 2023

BUTS Planet, Planet Campus

44

Pressure Drop in PBRs

$$\text{Concentration Flow System: } C_A = \frac{F_A}{v}$$

$$\text{Gas Phase Flow System: } v = v_0(1 + \varepsilon X) \frac{T}{T_0} \frac{P_0}{P}$$

$$C_A = \frac{F_A}{v} = \frac{F_{A0}(1-X)}{v_0(1+\varepsilon X) \frac{T}{T_0} \frac{P_0}{P}} = \frac{C_{A0}(1-X) \frac{T_0}{T} \frac{P}{P_0}}{(1+\varepsilon X)}$$

$$C_A = \frac{F_A}{v} = \frac{F_{A0} \left(\Theta_A - \frac{b}{a} X \right)}{v_0(1+\varepsilon X) \frac{T}{T_0} \frac{P_0}{P}} = \frac{C_{A0} \left(\Theta_A - \frac{b}{a} X \right) \frac{T_0}{T} \frac{P}{P_0}}{(1+\varepsilon X)}$$

29 September 2023

BUTS Planet, Planet Campus

45

PBR-2nd Order

Analyze the following second order gas phase reaction that occurs isothermally in a PBR:



Mole Balances

Must use the differential form of the mole balance to separate variables:

Rate Laws

Second order in A and irreversible:

$$F_{A0} \frac{dX}{dW} = -r'_A$$

$$-r'_A = k C_A^2$$

29 September 2023

MITS Plaza, Pinar Campus

46

Stoichiometry

$$C_A = \frac{F_A}{V} = C_{A0} \frac{(1-X)}{(1+\epsilon X)} \frac{P}{P_0} \frac{T_0}{T}$$

Isothermal, $T=T_0$

$$C_A = C_{A0} \frac{(1-X)}{(1+\epsilon X)} \frac{P}{P_0}$$

Combine:

$$\frac{dX}{dW} = \frac{k C_{A0}^2}{F_{A0}} \frac{(1-X)^2}{(1+\epsilon X)^2} \left(\frac{P}{P_0} \right)^2$$

Need to find (P/P_0) as a function of W (or V if you have a PFR)

29 September 2023

MITS Plaza, Pinar Campus

47

Ergun Equation: $\frac{dP}{dz} = \frac{-G}{\rho_0 G_r D_r \phi^3} \left[\frac{150(1-\phi)\mu}{D_r} + 1.75G \right] \frac{P}{P_0} \frac{T}{T_0} \frac{F_{T0}}{F_T}$

Constant mass flow: $\dot{m} = \dot{m}_0$

ϕ = porosity = $\frac{\text{volume of void}}{\text{total bed volume}}$ = void fraction

$\rho V = \rho_0 V_0$ $1 - \phi = \frac{\text{volume of solid}}{\text{total bed volume}}$

$\rho = \rho_0 \frac{V_0}{V}$ $\epsilon_0 = 32.174 \frac{\text{lb}_m}{\text{ft}^3} \cdot \text{ft}^3/\text{lb}_0 (\text{conversion factor})$

$= 4.17 \times 10^6 \frac{\text{lb}_m}{\text{ft}^3} \cdot \text{ft}^3/\text{lb}_0$

$$\rho = \rho_0 \frac{F_{T0}}{F_T} \frac{P_0}{P} \frac{T}{T_0}$$

$$V = V_0 (1 + \epsilon X) \frac{P_0}{P} \frac{T}{T_0}$$

29 September 2023

MITS Plaza, Pinar Campus

48

Variable Density $\rho = \rho_0 \frac{P}{P_0} \frac{T_0}{T} \frac{F_{T0}}{F_T}$

$$\frac{dP}{dz} = \frac{-G}{\rho_0 G_r D_r \phi^3} \left[\frac{150(1-\phi)\mu}{D_r} + 1.75G \right] \frac{P}{P_0} \frac{T}{T_0} \frac{F_{T0}}{F_T}$$

Let

$$\beta_0 = \frac{G}{\rho_0 G_r D_r \phi^3} \left[\frac{150(1-\phi)\mu}{D_r} + 1.75G \right]$$

29 September 2023

MITS Plaza, Pinar Campus

49



Catalyst Weight $W = zA_c \rho_c = zA_c (1 - \phi) \rho_c$

Where ρ_c = bulk density
 ρ_c = solid catalyst density
 ϕ = porosity (a.k.a., void fraction)
(1 - ϕ) = solid fraction

$$\frac{dP}{dW} = \frac{-\beta_c}{A_c (1 - \phi) \rho_c} \frac{P_c}{P} \frac{T}{T_0} \frac{F_r}{F_{T0}}$$

Let $\alpha = \frac{2\beta_c}{A_c (1 - \phi) \rho_c} \frac{1}{P_c}$



$$\frac{dp}{dW} = -\frac{\alpha}{2p} \frac{T}{T_0} \frac{F_r}{F_{T0}} \quad p = \frac{P}{P_0}$$

We will use this form for single reactions:

$$\frac{d(P/P_0)}{dW} = -\frac{\alpha}{2} \frac{1}{(P/P_0)} \frac{T}{T_0} (1 + \varepsilon X)$$

$$\frac{dp}{dW} = -\frac{\alpha}{2p} \frac{T}{T_0} (1 + \varepsilon X)$$

$$\frac{dp}{dW} = -\frac{\alpha}{2p} (1 + \varepsilon X) \quad \text{Isothermal case}$$



$$\frac{dX}{dW} = \frac{kC_{A0}^2 (1 - X)^2}{F_{T0} (1 + \varepsilon X)^2} p^2$$

$$\frac{dX}{dW} = f(X, p) \quad \frac{dP}{dW} = f(X, p) \quad \text{or} \quad \frac{dp}{dW} = f(p, X)$$

The two expressions are **coupled ordinary differential equations**. We can only solve them simultaneously using an ODE solvers.



$\varepsilon = 0$, analytical solution

For $\varepsilon = 0$

$$\frac{dp}{dW} = \frac{-\alpha}{2p} (1 + \varepsilon X)$$

When $W = 0 \quad p = 1$

$$dp^2 = -\alpha dW$$

$$p^2 = (1 - \alpha W)$$

$$p = (1 - \alpha W)^{1/2}$$

CHE 311 Kinetics & Reactor Design



Gas Phase Reaction in PBR for $\delta=0$

Gas Phase reaction in PBR with $\delta = 0$ (Analytical Solution)

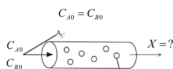


with equimolar feed of A and B and:

$$k_A = 1.5 \text{ dm}^3/\text{mol}/\text{kg}/\text{min}$$

$$\alpha = 0.0099 \text{ kg}^{-1}$$

Find X at 100 kg



29 September 2023

BTS-PHYS, PHEM CAMPUS

55

CHE 311 Kinetics & Reactor Design



1) Mole Balance

$$\frac{dX}{dW} = \frac{-r'_A}{F_{A0}}$$

2) Rate Law

$$-r'_A = k C_A C_B$$

3) Stoichiometry

$$C_A = C_{A0}(1-X)p$$

$$C_B = C_{A0}(1-X)p$$

29 September 2023

BTS-PHYS, PHEM CAMPUS

56

CHE 311 Kinetics & Reactor Design



Gas Phase Reaction in PBR for $\delta=0$

$$\frac{dp}{dW} = -\frac{\alpha}{2p} \quad 2pdp = -\alpha dW$$

$$W=0, \quad p=1 \quad p^2 = 1 - \alpha W$$

$$p = (1 - \alpha W)^{1/2}$$

4) Combine

$$-r'_A = k C_{A0}^2 (1-X)^2 p^2 = k C_{A0}^2 (1-X)^2 (1 - \alpha W)$$

$$\frac{dX}{dW} = \frac{k C_{A0}^2 (1-X)^2 (1 - \alpha W)}{F_{A0}}$$

29 September 2023

BTS-PHYS, PHEM CAMPUS

57

CHE 311 Kinetics & Reactor Design



$$\frac{dX}{(1-X)^2} = \frac{k C_{A0}^2}{F_{A0}} (1 - \alpha W) dW$$

$$\frac{X}{1-X} = \frac{k C_{A0}^2}{F_{A0}} \left(W - \frac{\alpha W^2}{2} \right)$$

$$W=0, X=0, W=W', X=X'$$

$$X = 0.6 \text{ (with pressure drop)}$$

$$X = 0.75 \text{ (without pressure drop, i.e. } \alpha = 0)$$

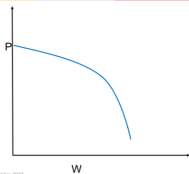
29 September 2023

BTS-PHYS, PHEM CAMPUS

58

CHE 311 Kinetics & Reactor Design

Pressure Drop in a PBR



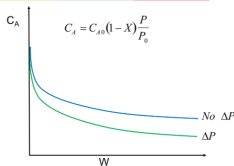
28 September 2023

BUTS Plains, Pinar Campus

59

CHE 311 Kinetics & Reactor Design

Concentration Profile in a PBR



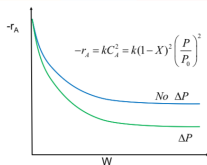
28 September 2023

BUTS Plains, Pinar Campus

60

CHE 311 Kinetics & Reactor Design

Reaction Rate in a PBR



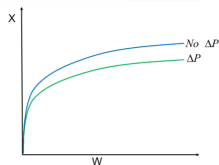
28 September 2023

BUTS Plains, Pinar Campus

61

CHE 311 Kinetics & Reactor Design

Conversion in a PBR



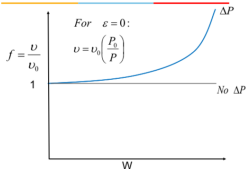
28 September 2023

BUTS Plains, Pinar Campus

62

CHE 331 Kinetics & Reactor Design

Flow Rate in a PBR



29 September 2023

MITS Place, Phase Campus

63

CHE 331 Kinetics & Reactor Design

PFR design with pressure drop

- Usual notation applies.
 - F_A = molar flow rate of A, F_T = total molar flow rate
 - V = volume of reactor, Q = volumetric flow rate
 - D = diameter and A = cross-sectional area of PFR, L = length of PFR, z = distance (between 0 and L)
 - T = temperature, R = universal gas constant, P = pressure, P_A = partial pressure of A
 - x = conversion, e = fractional increase in number of moles for 100% conversion, for a given feed conditions.
 - k = rate constant, C_A = concentration of A
 - ρ = density, μ = viscosity
 - f_f = friction factor, Re = Reynolds number
 - $-v_{a,p}$ = average velocity of the fluid in the PFR. Used sparingly.
 - Subscript "in" indicates inlet. E.g. $F_{A,in}$ is molar flow rate of A at the inlet.

29 September 2023

MITS Place, Phase Campus

69

CHE 331 Kinetics & Reactor Design

PFR design with pressure drop

Definition of ' x ' $F_A = F_{A-in}(1-x)$

Definition of ' e ' $F_T = F_{T-in}(1+ex)$

Definition of C_A $C_A = \frac{F_A}{Q}$

For gas phase only:

From ideal gas law $Q = \frac{F_T RT}{P} = \frac{F_{T-in}(1+ex)RT}{P}$

At constant temperature $Q = \frac{F_{T-in}(1+ex)RT}{P} = \frac{(1+ex)P_0 Q_0}{P}$

29 September 2023

MITS Place, Phase Campus

70

CHE 331 Kinetics & Reactor Design

For liquid as well as gas phase $Re = \frac{D(V_{a,p})\rho}{\mu} = \frac{4Q\rho}{\pi D\mu}$

$$f_f = \frac{16}{Re} \text{ if } Re < 2100$$

$$\approx \frac{1}{12.96 \left(\log_{10} \left(\frac{6.9}{Re} \right) \right)^2} \text{ if } Re > 10,000$$

29 September 2023

MITS Place, Phase Campus

71



• Steady state conditions $\frac{dF_A}{dV} = F_{A-0} = \frac{dx}{dV} = r_A$

• For a first order reaction $F_{A-0} \frac{dx}{dV} = k \frac{F_{A-0}(1-x)}{Q}$

$$F_{A-0} \frac{dx}{dV} = k \frac{F_{A-0}}{Q} = k \frac{F_{A-0}(1-x)}{Q}$$

• At constant temperature $\frac{dx}{dV} = k \frac{(1-x)}{Q} = \frac{k}{P_{in}Q_{in}} (1-x)P$

$$\frac{dx}{dz} = \frac{A}{P_{in}Q_{in}} k (1-x)P$$

29 September 2023

72

BTS Plots, Plots Campus

72



• For any other order of reaction also, write the equation such that $dx/dz = f(x, P)$. i.e. The only unknowns on the RHS must be x and P
 - i.e. RHS must not have Q or C_A . These are not known yet.
 - But Q_{in} and $C_{A, in}$ are known and hence RHS may have these terms.

$$\frac{dx}{dz} = \frac{A}{P_{in}Q_{in}} k (1-x)P = \beta \frac{(1-x)P}{(1+ex)}$$

where $\beta = \frac{A}{P_{in}Q_{in}} k C_{A, in}$

• For an 'n'th order reaction

$$\frac{dx}{dz} = k A \frac{F_{A-0}^{n-1}(1-x)^n P^n}{P_{in}^n Q_{in}^n (1+ex)^n} = \beta \frac{(1-x)^n P^n}{(1+ex)^n}$$

where $\beta = k A \frac{F_{A-0}^{n-1}}{P_{in}^n Q_{in}^n}$

29 September 2023

73

BTS Plots, Plots Campus



• Under steady state conditions, Reynolds number is a constant

- Even though local velocity changes
 - This is because density also changes with location
 - We assume that the viscosity of the medium remains the same, even when the reaction occurs

• Under steady state conditions, $(\rho Q) = \text{constant}$

• Using Re, Calculate the friction factor f_r .

• Write pressure drop equation: Hagen-Poiseuille

$$\frac{dP}{dz} = - \frac{32 f_r \rho Q}{\pi^2 D^3} = - \frac{32 f_r (\rho Q) Q}{\pi^2 D^3} = - \frac{32 f_r (\rho Q) Q}{\pi^2 D^3}$$

$$\frac{dP}{dz} = - \frac{32 f_r (\rho Q) Q}{\pi^2 D^3} = - \frac{32 f_r (\rho Q) Q}{\pi^2 D^3} (1+ex)$$

where $\alpha = \frac{32 f_r (\rho Q) Q}{\pi^2 D^3}$

29 September 2023

74

BTS Plots, Plots Campus

74



Solve both equations simultaneously

Initial conditions:

- At $z=0$, $P = P_{in}$
 - At $z=0$, $x = 0$

Special case:

- When $e=0$, solve the first equation and find 'P'. Then substitute for 'P' in the second equation and solve for 'x'.

$$\frac{dP}{dz} = - \alpha \frac{(1+ex)}{P}$$

where $\alpha = \frac{32 f_r (\rho Q) Q}{\pi^2 D^3}$

$$\frac{dx}{dz} = k A \frac{F_{A-0}^{n-1}(1-x)^n P^n}{P_{in}^n Q_{in}^n (1+ex)^n} = \beta \frac{(1-x)^n P^n}{(1+ex)^n}$$

where $\beta = k A \frac{F_{A-0}^{n-1}}{P_{in}^n Q_{in}^n}$

29 September 2023

75

BTS Plots, Plots Campus

CHE 311 Kinetics & Reactor Design



- The irreversible 1st order (wrt partial pressure of A) gas phase reaction $A \rightarrow B$ is carried out isothermally in a fluidized catalytic reactor containing 50 kg of catalysts. Currently 50% conversion is realized for pure A entering at pressure of 20 atm. There is virtually no pressure drop in CSTR. It is proposed to put a PBR containing the same catalyst weight in series with the CSTR. The pressure drop parameter for the PBR, α , is 0.018 kg^{-1} . The particle size is 0.2 mm, the bed porosity is 40%, and the viscosity is the same as that of air at 200 °C.
- a. Should the PBR be placed upstream or downstream of the CSTR in order to achieve the highest conversion?
- b. What is the conversion exiting the last reactor?
- c. What is the pressure at the exit of the packed bed?
- d. How would your answers change if the catalyst diameter were decreased by a factor of 2 and the PBR diameter were increased by 50% assuming turbulent flow?

29 September 2023

BUTS Pilon, Plain Campus

76

CHE 311 Kinetics & Reactor Design

RK-4th order

Runge-Kutta Method of 4th Order:

- Consider the differential Equation
 $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$
 Calculate successively
 - $k_1 = hf(x_0, y_0)$
 - $k_2 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1)$
 - $k_3 = hf(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2)$
 - $k_4 = hf(x_0 + h, y_0 + k_3)$
 Find $k = \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$
 $\therefore y_1 = y_0 + k$ and $x_1 = x_0 + h$

29 September 2023

BUTS Pilon, Plain Campus

77

CHE 311 Kinetics & Reactor Design

Batch Reactor



$A \rightarrow B \rightarrow C$, isothermal reaction takes place in a batch reactor. Where $k_1 = 1 \text{ hr}^{-1}$ and $k_2 = 2 \text{ hr}^{-1}$ and at time $t=0$, $C_A = 5 \text{ mol/m}^3$ and $C_B = C_C = 0 \text{ mol/m}^3$. Solve the system of equations and plot the change in concentration of each species over time.

```
function dcdt=batch(t,c)
c(1)=cA, c(2)=cB, c(3)=cC
global k1 k2
dcdt=[-k1*c(1); k1*c(1)-k2*c(2); k2*c(2)]

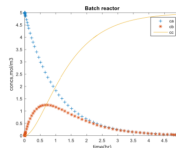
Command window
>> global k1 k2
>> k1=1; k2=2;
>> tspan=[0 5];
>> c0=[5; 0; 0];
>> [t,c]=ode45('batch',tspan,c0);
>> plot(t,c(:,1),'*','t,c(:,2),'*','t,c(:,3));
>> legend('cA','cB','cC');
>> xlabel('time(hr)');
>> ylabel('concn,mol/m3');
>> title('Batch reactor');
```

29 September 2023

BUTS Pilon, Plain Campus

78

CHE 311 Kinetics & Reactor Design

 $A \rightarrow B \rightarrow C$ 

29 September 2023

BUTS Pilon, Plain Campus

79

CHE 311 Kinetics & Reactor Design

1st Order ($A \rightarrow B$) – Plug flow

```
function dfunvecdv = PLUGQ(Vfunvec);
Ca = Vfunvec[1];
Cb = Vfunvec[2];
% Explicit equations
k = 0.23;
v0 = 10;
ra = -k * Ca;
rb = -ra;
% Differential equations
dCadv = ra / v0;
dCbdt = rb / v0;
dfunvecdv = [dCadv; dCbdt]; % Rate of the reaction along the length of the reactor
```

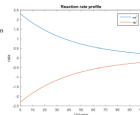
29 September 2023
BTS P/chem, P/chem Campus

80

CHE 311 Kinetics & Reactor Design

Rate@ along the length

```
clc
Vspan = [0 100]; % Range for the Volume of the reactor.
y0 = [10 0]; % Initial values for the dependent variables Ca and Cb
[cy] = ode45(@PLUG,Vspan,y0);
plot(cy);
k = 0.23;
for i = 1 : size(L,1)
    % Calculating the values of rate from its expression
    ra(i) = -k * cy(i,1);
    rb(i) = -ra(i);
end
plot(k, -ra, -rb)
xlabel('Volume')
ylabel('rate')
legend('-ra','rb')
title('Reaction rate profile')
```

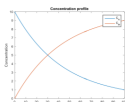

29 September 2023
BTS P/chem, P/chem Campus

81

CHE 311 Kinetics & Reactor Design

Concentration@length

```
clc
Vspan = [0 100]; % Range for the Volume of the reactor.
y0 = [10 0]; % Initial values for the dependent variables Ca and Cb
[cy] = ode45(@PLUG,Vspan,y0);
plot(cy(:,1),cy(:,2))
xlabel('Volume')
ylabel('Concentration')
legend('C_A','C_B')
title('Concentration profile')
```


29 September 2023
BTS P/chem, P/chem Campus

82

CHE 311 Kinetics & Reactor Design

PFR Design- MATLAB

- Determine the plug-flow reactor volume necessary to produce 300 million pounds of ethylene in a year from cracking a feed stream of pure ethane. The reaction is irreversible and follows an **elementary rate law**. We want to achieve **80% conversion** of ethane. Operating the reactor isothermally at 1100 K at a pressure of 6 atm. $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82 \text{ kJ/mol}$.

29 September 2023
BTS P/chem, P/chem Campus

83

CHE 311 Kinetics & Reactor Design



```
function dfunvecdV = ODEfun(Vfunvec);
X = Yfunvec(1);
% Explicit equations
k1 = 0.072;
Cao = 0.066;
T = 1100;
T1 = 1000;
R = 1.987;
E = 82000;
yao = 1;
Fao = 193;
epsilon = yao*(1+1-1);
k = k1*exp((E/R)*(1/T1-1/T));
Cb = Cao*X/(1+epsilon*X);
Cc = Cb;
Ca = Cao*(1-X)/(1+epsilon*X);
ra = -k*Cc;
% Differential equations
dXdV = - (ra / Fao);
dfunvecdV = [dXdV];
```

29 September 2023

BUTS Plots, Plant Campus

84

CHE 311 Kinetics & Reactor Design



Conversion Profile

clc

Vspan = [0 2310]; % Range for the Volume of the reactor.

y0 = [0]; % Initial values for the dependent variables X

[cy]=ode45(@ODEfun,Vspan,y0);

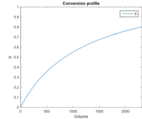
plot(cy(:,1])

xlabel('Volume')

ylabel('X')

legend('X')

title('Conversion profile')



29 September 2023

BUTS Plots, Plant Campus

85

CHE 311 Kinetics & Reactor Design



Concentration profile

clc

Vspan = [0 2310]; % Range for the Volume of the reactor.

y0 = [0]; % Initial values for the dependent variables X

[cy]=ode45(@ODEfun,Vspan,y0);

z=size(y);

Cao = 0.066;

yao=1;

epsilon = yao*(1+1-1);

for i=1 : z(1,1)

% Calculating the values of rate from its expression

Ca(i) = Cao*(1-y(i,1))/(1+epsilon*y(i,1));

Cb(i) = Cao*y(i,1)/(1+epsilon*y(i,1));

Cc(i) = Cb(i);

end

plot(cy,Ca,cy,Cb,cy,Cc)

xlabel('Volume')

ylabel('Concentration')

legend('C_A','C_B','C_C')

title('Concentration profile')

29 September 2023

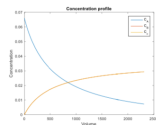
BUTS Plots, Plant Campus

86

CHE 311 Kinetics & Reactor Design



Concentration profile



29 September 2023

BUTS Plots, Plant Campus

87

CHE 311 Kinetics & Reactor Design

Rate profile

```

clc
Vspan = [0 2310]; % Range for the Volume of the reactor.
y0 = [0]; % Initial values for the dependent variables X
[cy]=ode45(@ODEfun,Vspan,y0);
r=isy(cy);
T = 1100;
T1 = 1000;
R = 1.987;
E = 82000;
k1 = 0.0372;
Cao = 0.056;
yao=1;
epsilon = yao*(1-1);
k = k1*exp((E/R)*(1/T1-1/T));
for i = 1: n(1,1)
    % Calculating the values of rate from its expression
    Ca(i) = Cao*(1-y(i,1))/(1+epsilon*y(i,1));
    ra(i) = -k*Ca(i);
    and
    plot(V,-ra)
    xlabel('Volume')
    ylabel('-ra^2')
    legend('rate')
    title('Reaction rate profile')

```

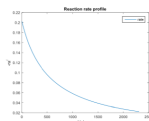
29 September 2023

BUTS Planet, Planet Campus

88

CHE 311 Kinetics & Reactor Design

Reaction rate profile



29 September 2023

BUTS Planet, Planet Campus

89

CHE 311 Kinetics & Reactor Design

PFR- pressure drop

- Consider a gas phase reaction under isothermal conditions. (Isomerization)

A → B

$P_{in} = 10$ atm, $Q = 0.005$ m³/s, $T = 300$ K, Pure A is fed, $k = 0.1$ lit/s, Molecular weight $M = 60$ g/mol, Viscosity of the gas $= 10^{-5}$ Pa-s

What should be the length of the PFR if it is constructed (a) using a 2 cm pipe and the conversion desired is 10%? (b) using a 1.5 cm pipe and the conversion desired is 10% and (c) using a 1.5 cm pipe and the conversion desired is 20%

29 September 2023

BUTS Planet, Planet Campus

90

CHE 311 Kinetics & Reactor Design

c^o = 0. This simplifies the equations

$$\frac{dP}{dV} = -\frac{P}{V} \left(\frac{1+x}{P} \right)$$

$$\frac{dP}{dV} = -\frac{1}{V} \quad P^2 = P_{in}^2 - 2\alpha x$$

R=0.08206 atm-lit/(gmol-K)

$$\rho_{in} = \frac{P_{in} M}{RT} = 24.0558 \text{ kg/m}^3$$

$$F_{Ain} = 2.0047 \text{ mol/s}$$

$$V_{Ain} = 15.9155 \text{ m}^3/\text{s}$$

$$Re = 7.65 \times 10^5$$

$$f = 0.003$$

$$\alpha = 1.847 \times 10^5 \text{ Pa}^2/\text{m}$$

$$\beta = 6.2832 \times 10^{-4} \text{ 1/(Pa m)}$$

$$\frac{dx}{dV} = k \frac{P_{in}^2 - (P - \beta x)^2}{P_{in}^2 (1 + \alpha x)} = \beta \frac{(1 - \beta x)^2}{(1 + \alpha x)^2}$$

where $\beta = k \frac{P_{in}^2}{P_{in}^2}$

29 September 2023

BUTS Planet, Planet Campus

91

CHE 333 Kinetics & Reactor Design



```
% Differential equation that goes with pressure.drop2.m
%%
function dY = dYdx(x,Y,p1)

dY = zeros(2,1);
% Y(1) is pressure, Y(2) is conversion
alpha = p1(1); beta = p1(2); epsilon = p1(3); n = p1(4);
P = Y(1); X = Y(2);

dPdZ = -alpha*(1+epsilon * X) / P;
dXdZ = beta*(1-X)^n * P^n / (1+epsilon * X)^n;

dY(1) = dPdZ; dY(2) = dXdZ;
dY;
end
```

29 September 2023

BTS Plots, Pisan Campus

92

CHE 333 Kinetics & Reactor Design



```
clear all;
close all;

Pin = 1e6; %Pa;
T = 300; %Kelvin
Qin = 0.005; % m^3/s;
M = 60; % molecular weight, g/mol
dia = 0.02; % diameter in m
mu = 1e-5; % viscosity in Pa-s
k = 0.1; % rate constant with correct units, 1/s, for example.
n = 1; % order of reaction
epsilon = 0; % fractional change in volume for 100% conversion, for the given feed
z_end = 20; % in meters
% Integrate from 0 to some value. Hopefully, the desired conversion will be
% within this. Else, integrate for longer distance and check.

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Input values are given above
xin = 0; % conversion at inlet, usually taken as zero.
```

29 September 2023

BTS Plots, Pisan Campus

93

CHE 333 Kinetics & Reactor Design



```
R = 8.314; %Pa-m^3/gmol/K
RhoIn = Pin * M/(R*T)/1000; % density at inlet, kg/m^3
Fain = Pin * Qin / (R*T); % molar flow rate of A at inlet, gmol/s
area = pi * dia^2 / 4;
Vain = Qin/area;

Re = dia * Vain * RhoIn/mu;

fflnv = 12.96 * (log10(6.9/Re))^2;
ff = 1/fflnv;

alpha = 32 * ff*(RhoIn * Qin)*(Pin * Qin)/(pi^2 * dia^5)
beta = k * area * Fain^(n-1)/(Pin*Qin)^n

% Now, the two differential equations must be solved simultaneously.
%
p1(1)=alpha; p1(2)=beta; p1(3)=epsilon; p1(4)=n;
Y_init = [Pin xin];
```

29 September 2023

BTS Plots, Pisan Campus

94

CHE 333 Kinetics & Reactor Design



```
options = odeset('RelTol',1e-4,'AbsTol',1e-4);
[z,Y] = ode15s(@x,ydYdx,x,p1,[0 z_end],Y_init,options);

Pressure = Y(:,1); Conversion=r(:,2);
%% plot the results

figure;
axes('Position',[0.15 0.15 0.75 0.3]);
fig1 = plot(z,Pressure); hold on;
set(fig1,'Color',[1 0 0],'Marker','o','MarkerSize',8,...
'MarkerEdgeColor',[1 0 0],'MarkerFaceColor',[1 1 1],...
'LineWidth',1,'LineStyle','-');
xlabel('PFR length / m','fontSize',14);
ylabel('Pressure / Pa','fontSize',14);

text(0.2,0.3,['epsilon = ' num2str(epsilon)],'Units','normalized');
text(0.2,0.4,['d = ' num2str(dia*100) ' cm'],'Units','normalized');
```

29 September 2023

BTS Plots, Pisan Campus

95

29-09-2023

CHE 551 Kinetics & Reactor Design



```
axes[Position],[0.15 0.6 0.75 0.3]);  
fig2 = plot(x,Conversion);  
  
set(fig2,'Color',[0 0 1],'Marker','*','MarkerSize',8,...  
    'MarkerEdgeColor',[0 0 1],'MarkerFaceColor',[1 1 1],...  
    'LineWidth',1,'LineStyle','-');  
  
ylabel('Conversion','fontSize',14);  
title('Conversion and Pressure vs distance','fontSize',14);  
  
text(0.2,0.3,['epsilon = ' num2str(epsilon)],'Units','normalized');  
text(0.2,0.4,['d = ' num2str(dia*100) ' cm'],'Units','normalized');  
  
set(gcf,'color',[1 1 1]);
```

29 September 2023

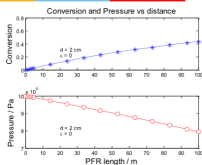
B7S Plots, Plot Compare

96

CHE 551 Kinetics & Reactor Design



Dia = 2 cm



29 September 2023

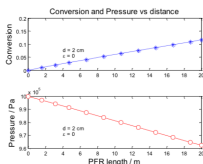
B7S Plots, Plot Compare

97

CHE 551 Kinetics & Reactor Design



Dia = 2 cm, expanded



29 September 2023

B7S Plots, Plot Compare

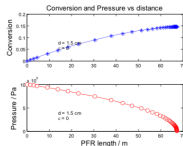
98

CHE 551 Kinetics & Reactor Design



Dia = 1.5 cm

10% conversion is possible, but 20% is not



29 September 2023

B7S Plots, Plot Compare

99